This document has been translated from Spanish to English.

The data have been anonymized.

Participant flow

Of the 286 older adults invited to participate, 262 agreed to take part in the evaluation. A total of 250 participants completed the post-use questionnaire and were included in the final analysis.

Stage	*n*	*%*
Invited to participate	**286**	**100.0**
Agreed to participate	**262**	**91.6**
Completed the trial period	**254**	**88.8**
Completed the final survey	**250**	**87.4**
Withdrew before final survey	**12**	**4.2**
Excluded because of incomplete responses	**4**	**1.4**

Demographic profile

Variable	*Category*	*n*	*%*
Age	65-74	96	38.4
	75-84	111	44.4
	85+	43	17.2
Gender	Women	151	60.4
	Men	97	38.8
	Other / prefer not to say	2	0.8
Living situation	Alone	128	51.2
	With partner	74	29.6
	With family	34	13.6
	Assisted/community housing	14	5.6
Education	Primary or less	104	41.6
	Secondary	91	36.4
	Vocational/university	43	17.2
	Prefer not to say	12	4.8
Digital literacy	Low or none	84	33.6
	Basic	114	45.6
	Regular digital-device use	52	20.8
Previous voice-assistant use	None	164	65.6
	Occasional	62	24.8
	Regular	24	9.6
Care-support level	No formal care	33	13.2
	Occasional support	70	28.0
	Regular home-care support	108	43.2
	High-dependence support	39	15.6

Trial exposure

Most participants used the device daily or several times per week, suggesting that the post-use survey reflected practical exposure rather than a single demonstration session.

Variable	***Result***
Mean number of trial days	21.4 days
Standard deviation	5.7 days
Median number of trial days	22 days
Participants using SAM daily	152 / 250, 60.8%
Participants using SAM several times per week	71 / 250, 28.4%
Participants using SAM once per week	20 / 250, 8.0%
Participants using SAM less than once per week	7 / 250, 2.8%

Assistance needed

Assistance during use	***n***	***%***
Never needed help	86	34.4
Needed help once or twice	82	32.8
Needed help sometimes	58	23.2
Needed help often	24	9.6

Item-level results

Perceived usefulness

Item	*Statement*	*Agree / strongly agree*
PU1	SAM was useful in my daily routine.	213 / 250, 85.2%
PU2	SAM helped me feel more accompanied at home.	201 / 250, 80.4%
PU3	SAM could help me remember important activities.	224 / 250, 89.6%
PU4	SAM made it easier to ask for information or assistance.	208 / 250, 83.2%
PU5	SAM could complement support from family or care professionals.	228 / 250, 91.2%

Ease of use

Item	*Statement*	*Agree / strongly agree*
EU1	It was easy to start using SAM.	205 / 250, 82.0%
EU2	I knew what to say to SAM.	191 / 250, 76.4%
EU3	Speaking to SAM was easier than using a screen or keyboard.	221 / 250, 88.4%
EU4	I could use SAM without needing much help.	186 / 250, 74.4%
EU5	The interaction with SAM felt simple and clear.	207 / 250, 82.8%

Voice interaction quality

Item	*Statement*	*Agree / strongly agree*
VQ1	SAM understood my voice correctly most of the time.	184 / 250, 73.6%
VQ2	SAM's answers were easy to understand.	214 / 250, 85.6%
VQ3	SAM responded quickly enough.	192 / 250, 76.8%
VQ4	SAM's voice was pleasant or acceptable.	218 / 250, 87.2%
VQ5	When SAM made a mistake, it was easy to try again.	176 / 250, 70.4%

Trust, privacy and safety

Item	*Statement*	*Agree / strongly agree*
TS1	I felt safe using SAM.	221 / 250, 88.4%
TS2	I trusted SAM to respond appropriately.	203 / 250, 81.2%
TS3	I was comfortable having SAM in my home.	216 / 250, 86.4%
TS4	I understood when SAM was listening.	197 / 250, 78.8%
TS5	I was worried about my privacy.	47 / 250, 18.8%
TS6	I was afraid of making mistakes when speaking to SAM.	53 / 250, 21.2%

Overall acceptance rate

A participant was classified as showing positive acceptance if they answered “agree” or “strongly agree” to “I would like to continue using SAM if it were available” and obtained a mean score of at least 4.0 across the overall acceptance items.

Positive acceptance was observed in 217 of the 250 valid responses, corresponding to an overall acceptance rate of 86.8% 95% CI: 82.6-91.0. The mean overall acceptance score was 4.34 out of 5 SD = 0.59, indicating high acceptance of the system after the trial period.

Measure	***Result***
Participants meeting acceptance criterion	217 / 250
Overall acceptance rate	86.8%
Approximate 95% confidence interval	82.6% to 91.0%
Mean overall acceptance score	4.34 / 5
SD	0.59
Median	4.50 / 5

Subgroup results

Acceptance was high across all age groups, although it was lower among participants aged 85 or older. Participants with regular digital-device use and those with previous experience using voice assistants showed the highest acceptance. However, acceptance among participants with low or no digital literacy remained substantial at 79.8%, suggesting that the voice-based interaction reduced some barriers associated with screen-based interfaces.

Subgroup	***Accepted / total***	***Acceptance rate***
Age 65-74	87 / 96	90.6%
Age 75-84	96 / 111	86.5%
Age 85+	34 / 43	79.1%
Low/no digital literacy	67 / 84	79.8%
Basic digital literacy	100 / 114	87.7%
Regular digital-device use	50 / 52	96.2%
No previous voice-assistant use	136 / 164	82.9%
Occasional previous use	57 / 62	91.9%
Regular previous use	24 / 24	100.0%
Living alone	114 / 128	89.1%
Not living alone	103 / 122	84.4%

Open-ended responses

Theme	***Frequency***	***Interpretation***
Useful for reminders and daily routines	96 / 250, 38.4%	Participants valued practical support
Feeling accompanied	83 / 250, 33.2%	The system was perceived as socially useful
Easier than screen-based devices	67 / 250, 26.8%	Voice interaction reduced interface barriers
Speech recognition problems	61 / 250, 24.4%	Main technical limitation
Privacy/listening concerns	42 / 250, 16.8%	Important deployment issue
Desire for more natural conversation	39 / 250, 15.6%	Future improvement for dialogue model
Need for human backup	73 / 250, 29.2%	Users saw SAM as complementary, not a replacement

Conclusions

A total of 250 older adults completed the post-use acceptance questionnaire. Participants were mostly aged between 75 and 84 years 44.4%, and 51.2% lived alone. Most participants had either low/no digital literacy 33.6% or basic digital literacy 45.6%, and 65.6% had no previous experience using voice assistants. The mean trial exposure was 21.4 days SD = 5.7, and 89.2% of participants used the device daily or several times per week.

The highest-rated dimensions were overall acceptance M = 4.34, SD = 0.59 and perceived usefulness M = 4.31, SD = 0.62. Voice interaction quality obtained a slightly lower score M = 3.94, SD = 0.76, mainly because some participants reported occasional recognition errors or delays. Internal consistency was acceptable to high across the main scales, with Cronbach's alpha ranging from 0.79 to 0.88.

Positive acceptance was defined as agreement or strong agreement with the intention-to-continue item, together with a mean score of at least 4.0 across the overall acceptance items. According to this criterion, 217 of the 250 participants showed positive acceptance, corresponding to an acceptance rate of 86.8% 95% CI: 82.6-91.0. Acceptance remained high across age groups, although it was lower among participants aged 85 years or older. Perceived usefulness and voice interaction quality were the strongest predictors of positive acceptance, while privacy concern was negatively associated with acceptance.

Open-ended responses indicated that participants especially valued reminders, companionship, and the possibility of interacting without a screen. The most frequent limitations concerned occasional speech-recognition errors, uncertainty about when the device was listening, and the need to ensure that the system complements rather than replaces human care.